

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate Investigational Agents With or Without Pembrolizumab (MK-3475) in Participants With Advanced Esophageal Cancer Previously Exposed to Programmed Cell Death 1 Protein (PD-1)/ Programmed Cell Death Ligand 1 (PD-L1) Treatment (MK-3475-06B) **Official Title:** A Phase 1/2 Open-Label, Umbrella Platform Design Study of Investigational Agents With or Without Pembrolizumab (MK-3475) and/or Chemotherapy in Participants With Advanced Esophageal Cancer Previously Exposed to PD-1/PD-L1 Treatment (KEYMAKER-U06): Substudy 06B **Short Title/Acronym:** KEYMAKER-U06B / MK-3475-06B **Protocol Number:** MK-3475-06B **NCT Number:** NCT05319730 **Posting ID:** 633043043474438080 **Sponsor/ Company Name:** Merck **Investigational Product:** Pembrolizumab (MK-3475) combined with Investigational Agents (e.g., MK-4830) +/- Chemotherapy **Phase of Development:** PHASE1, PHASE2 **Trial Status:** RECRUITING **Medical Monitor:** Merck Clinical Director / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and efficacy (assessing Dose-Limiting Toxicities [DLTs] and Adverse Events [AEs] during safety lead-in phases) of investigational agents with or without pembrolizumab and/or chemotherapy in participants with advanced esophageal squamous cell carcinoma (ESCC). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS), and Duration of Response (DOR).

2.2 Background & Rationale To address the unmet medical need for patients with second-line (2L) esophageal squamous cell carcinoma (ESCC) who have experienced disease progression. Advanced ESCC patients who have already been exposed to first-line standard therapy (including a platinum agent and anti-PD-1/PD-L1 immune oncology therapy) face limited effective treatment options. This umbrella study evaluates novel immunotherapy and chemotherapy combinations to overcome resistance and improve survival outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Standard of Care (dependent on specific platform arm) **Blinding:** Open-label **Randomization:** Randomized (Umbrella Platform Design)

3.2 Planned Enrollment & Duration Targeted Enrollment: 230 participants **Number of Sites:** Multicenter (Global, including US, EU, and Japan) **Estimated Study Start:** 16-May-2023 **Estimated Primary Completion:** 11-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of metastatic or locally advanced unresectable Esophageal Squamous Cell Carcinoma (ESCC). 3. Has experienced investigator documented radiographic or clinical disease progression on one prior line of standard therapy, that includes a platinum agent and previous exposure to an anti-programmed cell death 1 (PD-1)/programmed cell death ligand 1 (PD-L1) based immune oncology (IO) therapy. 4. Has provided an archival or most recent tumor tissue sample obtained as part of clinical practice. 5. Participants who have adverse events (AEs) due to previous anticancer therapies must have recovered to $\leq$ Grade 1 or baseline. Participants with endocrine-related AEs who are adequately treated with hormone replacement or participants who have $\leq$ Grade 2 neuropathy are eligible.

4.2 Exclusion Criteria 1. Direct invasion into adjacent organs such as the aorta or trachea. 2. Has experienced weight loss $> 10\%$ over approximately 2 months prior to the first dose of study therapy. 3. Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 4. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 5. Has received an investigational agent or has used an investigational device within 4 weeks prior to study intervention administration. 6. Known additional malignancy that is progressing or has required active treatment within the past 3 years (with specific exceptions). 7. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 8. Participants with human immunodeficiency virus (HIV) with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 9. History of allogenic tissue/solid organ transplant. 10. Clinically significant cardiovascular disease within 12 months from first dose. 11. Has risk for significant gastrointestinal (GI) bleeding or has had a significant bleeding episode from the GI tract within 12 weeks prior to allocation/randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab administered intravenously (IV) alongside investigational agents (and/or chemotherapy) at protocol-specified intervals per

specific treatment arm. **Study Arms Update:** As of Protocol Amendment 5, the Pembrolizumab Plus MK-4830 Plus Paclitaxel/Irinotecan arm and the Pembrolizumab Plus MK-4830 Plus Lenvatinib arm are no longer actively enrolling participants. **Route of Administration:** Intravenous (IV) and/or Oral (dependent on the specific investigational agent/chemotherapy arm). **Duration of Treatment:** Up to 2 years (or until disease progression, unacceptable toxicity, or patient withdrawal).

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care, hormone replacement therapy for endocrine-related AEs, and standard analgesics including paracetamol (ATC Code: N02BE01). **Prohibited:** Other investigational agents/devices within 4 weeks prior to study start, systemic immunosuppressive therapies, and live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue sample provision, Baseline Scans
Baseline	Day 0	Randomization, First Dose of Study Treatment, Safety Labs
Interim	Every 3 to 6 Weeks	Safety Labs, Efficacy Assessment (Tumor Imaging), AE monitoring, Dosing
End of Study	Up to 2 Years	Final Vital Signs, Exit Scans, Survival Follow-up, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants Experiencing Dose-Limiting Toxicities (DLTs) During Safety Lead-in Phase:** A DLT is defined as any drug-related AE according to NCI CTCAE Version 5.0 that results in a dose change or delay.
- 2. Number of Participants Who Experienced an Adverse Event (AE) During Safety Lead-in Phase:** Any untoward medical occurrence temporally associated with the use of study intervention.
- 3. Number of Participants Who Discontinue Study Treatment Due to an AE During Safety Lead-in Phase.**

7.2 Secondary & Exploratory Endpoints

- 1. Progression-Free Survival (PFS)**
- 2. Duration of Response (DOR)**
- 3. Overall Survival (OS)**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and DLTs during the Phase 1 portion and throughout Phase 2, evaluated via physical exams, laboratory tests, and vital signs. **Serious Adverse Events (SAEs):** Standard reporting timelines strictly adhered to (typically within 24 hours of investigator awareness).

Data Monitoring Committee (DMC): An independent Data Monitoring Committee (iDMC) assesses safety and efficacy signals across the umbrella platform arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set (all patients receiving ≥ 1 dose) for safety/DLT analysis. **Sample Size Justification:** Targeted Enrollment = 230, powered to detect clinically meaningful improvements compared to historical controls for 2L ESCC. **Handling of Missing Data:** Standard censoring rules for time-to-event endpoints (Kaplan-Meier estimates for PFS and OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Validated Electronic Data Capture (EDC) system with strict access control, full audit trails, and independent central imaging reviews.

11. Study Identifiers & Governance

Primary ID: MK-3475-06B **Registry Number:** NCT05319730 **Posting ID:** 633043043474438080 **Sponsor/Lead Organization:** Merck **Study Title:** KEYMAKER-U06B: Pembrolizumab Combinations in 2L+ ESCC

12. Clinical Framework (Disease Specific)

Disease Areas: Esophageal Squamous Cell Carcinoma **Preferred Disease Name:** Metastatic Esophageal Squamous Cell Carcinoma **Preferred UMLS Name:** Esophageal Neoplasms, Drug-Resistant **Semantic Type:** Neoplastic Process **Definition:** A squamous cell carcinoma that arises from the esophagus and has metastasized to another anatomic site. **MeSH Code:** C04.588.322.250 **SNOMED ID:** 254636009 **Diagnosis Threshold:** Documented progression on a prior line of platinum-based chemotherapy and PD-1/PD-L1 inhibitor. **Medical Methodology:** Immunotherapy combination regimens +/- Cytotoxic Chemotherapy.

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced ESCC, integrating investigational targeted therapies.

Education Protocol (HCP): Site initiation visits covering protocol specifics, DLT management, and RECIST 1.1 evaluation.
Education Protocol (Patient): Informed consent review, treatment schedule familiarization, and AE reporting instructions. **Quality Audit Mechanism:** Source data verification, routine site audits, and centralized, blinded independent review of radiographic images.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or study completion (up to approximately 2 years of active treatment). **Onsite Visit Cadence:** Aligned with treatment cycles. **Remote Monitoring Frequency:** Continuous survival follow-up post-treatment discontinuation. **Checklist Review Interval:** Tumor imaging assessments conducted every 9 weeks (Q9W) for the first year, then every 12 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				